5.5.1 Military and defense applications

The military remains the primary arena where fourth-generation drones have achieved the greatest impact. Equipped with AI-assisted autonomy, stealth design, and multi-sensor fusion, these UAVs are redefining modern warfare and intelligence operations through their capacity for precision, persistence, and adaptive decision-making.

- *Intelligence, surveillance, and reconnaissance (ISR)*: Drones like the RQ-180 and Heron TP employ multi-spectral imaging, SAR radars, and ELINT payloads for persistent surveillance over hostile territories, offering unmatched situational awareness.
- *Autonomous combat and strike missions*: Platforms such as the MQ-25 Stingray and Loyal Wingman execute coordinated combat and refueling operations autonomously, enabling extended missions and reducing human risk in high-threat environments.
- *Electronic and cyber warfare*: Fourth-generation UAVs integrate electronic attack (EA) and electronic support (ES) capabilities for radar jamming, signal deception, and communications disruption, contributing to information dominance in the digital battlespace.
- *Swarm and cooperative engagement*: AI-driven swarms perform saturation strikes, overwhelm enemy air defenses, and conduct multi-directional intelligence gathering, enhancing both offensive and defensive mission outcomes.
- *Logistical and tactical support*: Heavy-lift drones and hybrid UAVs deliver supplies, ammunition, and medical aid in battlefield zones, ensuring rapid resupply and casualty evacuation in environments inaccessible to traditional vehicles.